

DATA STRUCTURE AND ALGORITHM USING PYTHON

SOCIAL MEDIA MARKETING AGENCY SYSTEM

(Using Linked List and File Handling)

INTRODUCTION

A **social media marketing agency** is a business that helps other businesses promote their products and services on social media platforms such as Instagram, Facebook, YouTube and LinkedIn. The agency designs posts, edits reels and short videos, runs paid advertising campaigns and manages the social media accounts of its clients. Every client chooses a platform, a service and a monthly budget, so the agency has to keep an accurate record of all its clients.

The **Social Media Marketing Agency System** is a Python program that keeps this client record. It uses a **file** to store the data, so the records are not lost when the program is closed. The program provides the following functions:

1. You can **add** new client records.
2. It will **display** all the client records stored already, with the total monthly budget.
3. The **search** function finds any stored client by name or phone number.
4. You can **update** a client's service package and budget.
5. You can **delete** a client record.

Before using the system, the staff member must log in. The username and password are checked against the file **login.txt**.

The client list is created using the **Linked List** data structure of Python. A linked list provides a logical connection between data elements that are stored in memory at different locations physically. Data elements are stored in **nodes** along with a reference **link** to the next data element. The first node is called the **head** and the link of the last node is **None**. The entire list is accessed sequentially by starting from the head and moving from one node to the next using these links.



Figure 1: Client records stored as nodes of a singly linked list.

SWOT ANALYSIS

SWOT analysis looks at the **Strengths** and **Weaknesses** inside a business and the **Opportunities** and **Threats** outside it. The SWOT analysis of a social media marketing agency is given below.

S STRENGTHS

- Low start-up cost – needs only computers, internet and a skilled team.
- Can reach a large and exactly targeted audience by age, city and interests.
- Results are measurable – reach, likes, clicks and leads can be shown to clients.
- Many services in one place: post design, reels, ads and account management.
- Monthly packages give the agency a regular income.

W WEAKNESSES

- Depends on platforms it does not control; an algorithm change can cut reach overnight.
- Needs fresh creative content every day, which is hard to keep up.
- Clients expect quick results, but building a brand takes time.
- A small agency depends on a few clients; losing one affects income.
- Client records kept on paper or in chats lead to mistakes and missed follow-ups.

O OPPORTUNITIES

- Small and local businesses (shops, cafés, gyms) are moving online.
- Short videos (Reels, Shorts) and influencer marketing are growing fast.
- Regional-language content for smaller cities and towns.
- AI tools for designs, captions and scheduling save time.
- Online stores need paid ads to get sales.

T THREATS

- Heavy competition from other agencies and low-cost freelancers.
- Businesses can use free tools (Canva, Meta Business Suite) and do it themselves.
- Sudden changes in platform rules or advertising prices.
- Data privacy laws limit how customer data can be used for targeting.
- In a slowdown, marketing budgets are the first to be cut.

LEARNING FROM THE CASE

From the social media marketing agency case

1. An agency's biggest asset is its client list. Every client's platform, service and budget must be recorded correctly and be easy to find.
2. Depending on one platform or one big client is risky; the agency should spread its work across platforms and clients.
3. Showing clear, measurable results (reach, leads, sales) keeps clients coming back every month.
4. The market changes quickly – reels, influencers, AI tools, new rules – so the agency must keep learning and adapting.
5. Knowing the total monthly budget of all clients helps the agency plan its staff and income.

From building the system

6. A linked list grows as clients are added. There is no fixed size to decide in advance, unlike an array.
7. Display, search, update and delete all work by **traversing** the list from the head, one node at a time. Searching this way is called **linear search**.
8. Deleting a client only changes one link (`previous.next = current.next`). No elements have to be shifted as in an array.
9. **File handling** makes the data permanent: the list is built from `clients.txt` when the program starts and saved back to the file after every change.
10. Checking the login from `login.txt` is better than writing usernames and passwords inside the code.
11. Checking input (10-digit phone, number for budget, menu choices) prevents wrong data from being stored.

PROGRAM DESIGN

Files used

File	What it stores	Format of one line
<code>clients.txt</code>	Client records (data)	<code>name,phone,platform,service,budget</code>
<code>login.txt</code>	Staff login details	<code>username,password</code>

Linked list operations used

Operation	Function	How it works
Insert at end	<code>add_client()</code>	Move to the last node and link the new node after it
Traversal	<code>display_clients()</code>	Visit every node from head to end; print, count and add budgets
Linear search	<code>find_client()</code> <code>search_client()</code>	Compare each node's name / phone until a match is found
Update	<code>update_client()</code>	Find the node and change its service and budget
Delete	<code>delete_client()</code>	Link the previous node to the next node, skipping the deleted one
Save / Load	<code>save_to_file()</code> <code>load_from_file()</code>	Write each node as one line / build the list from the lines

SOURCE CODE:-

agency.py

```
# SOCIAL MEDIA MARKETING AGENCY SYSTEM
# Data Structure used : Singly Linked List
# Files used          : clients.txt -> client records (name,phone,platform,service,budget)
#                     login.txt   -> staff login details (username,password)

CLIENT_FILE = "clients.txt"
LOGIN_FILE = "login.txt"

PLATFORMS = ["Instagram", "Facebook", "YouTube", "LinkedIn"]
SERVICES = ["Post Design", "Reels Editing", "Ad Campaign", "Account Management"]

class Client:
    # one node of the linked list = one client of the agency
    def __init__(self, name, phone, platform, service, budget):
        self.name = name
        self.phone = phone
        self.platform = platform
        self.service = service
        self.budget = budget
        self.next = None

class AgencyList:
    def __init__(self):
        self.head = None

    # insert a new client node at the end of the list
    def add_client(self, name, phone, platform, service, budget):
        new_node = Client(name, phone, platform, service, budget)
        if self.head is None:
            self.head = new_node
            return
        current = self.head
        while current.next is not None:
            current = current.next
        current.next = new_node

    # visit every node from head to the end and print it
    def display_clients(self):
        if self.head is None:
            print("--- No clients available ---")
            return
        print(f"{'NO':<4}{'NAME':<19}{'PHONE':<12}{'PLATFORM':<11}{'SERVICE':<20}{'BUDGET':>8}")
        print("-" * 74)
        count = 0
        total = 0
        current = self.head
        while current is not None:
            count = count + 1
            total = total + current.budget
            print(f"{count:<4}{current.name[:18]:<19}{current.phone:<12}{current.platform:<11}"
                  f"{current.service:<20}{current.budget:>8}")
            current = current.next
        print("-" * 74)
        print("Total Clients :", count, "      Total Monthly Budget : Rs.", total)

    # linear search by name or phone number, returns the node or None
    def find_client(self, key):
        current = self.head
        while current is not None:
            if current.name.lower() == key.lower() or current.phone == key:
                return current
            current = current.next
        return None

    def search_client(self, key):
        node = self.find_client(key)
        if node is None:
```

```

        print("Sorry!", key, "is not available in the client list")
    else:
        print("Client Found!")
        print("Name      :", node.name)
        print("Phone     :", node.phone)
        print("Platform  :", node.platform)
        print("Service   :", node.service)
        print("Budget    : Rs.", node.budget)

def update_client(self, key, new_service, new_budget):
    node = self.find_client(key)
    if node is None:
        print("Sorry!", key, "is not available in the client list")
        return False
    node.service = new_service
    node.budget = new_budget
    print(node.name, "updated successfully!")
    return True

# remove a node by changing the link of the previous node
def delete_client(self, key):
    if self.head is None:
        print("--- No clients available ---")
        return False
    if self.head.name.lower() == key.lower() or self.head.phone == key:
        print(self.head.name, "deleted successfully!")
        self.head = self.head.next
        return True
    previous = self.head
    current = self.head.next
    while current is not None:
        if current.name.lower() == key.lower() or current.phone == key:
            previous.next = current.next
            print(current.name, "deleted successfully!")
            return True
        previous = current
        current = current.next
    print("Sorry!", key, "is not available in the client list")
    return False

# write every node to the file, one client per line
def save_to_file(self):
    with open(CLIENT_FILE, "w") as file:
        current = self.head
        while current is not None:
            file.write(f"{current.name},{current.phone},{current.platform},"
                      f"{current.service},{current.budget}\n")
            current = current.next

# read the file and build the linked list
def load_from_file(self):
    try:
        with open(CLIENT_FILE, "r") as file:
            for line in file:
                parts = line.strip().split(",")
                if len(parts) == 5:
                    self.add_client(parts[0], parts[1], parts[2], parts[3], int(parts[4]))
    except FileNotFoundError:
        print("(clients.txt not found - starting with an empty list)")

# check the username and password against login.txt
def check_login(username, password):
    try:
        with open(LOGIN_FILE, "r") as file:
            for line in file:
                parts = line.strip().split(",")
                if len(parts) == 2 and parts[0] == username and parts[1] == password:
                    return True
    except FileNotFoundError:
        print("login.txt not found!")
    return False

```

```

def choose(title, options):
    for i in range(len(options)):
        print(f" {i + 1}. {options[i]}")
    while True:
        pick = input(title)
        if pick.isdigit() and 1 <= int(pick) <= len(options):
            return options[int(pick) - 1]
        print("Please enter a number from 1 to", len(options))

def ask_budget():
    while True:
        budget = input("Monthly Budget (Rs.) : ")
        if budget.isdigit() and int(budget) > 0:
            return int(budget)
        print("Budget must be a number greater than 0")

# ----- MAIN PROGRAM -----
print("=" * 50)
print("    SOCIAL MEDIA MARKETING AGENCY SYSTEM")
print("=" * 50)
user_name = input("Hello! Welcome --- What's your name : ")

logged_in = False
attempts = 3
while attempts > 0 and not logged_in:
    username = input(user_name + ", please enter Username : ")
    password = input(user_name + ", please enter Password : ")
    if check_login(username, password):
        logged_in = True
    else:
        attempts = attempts - 1
        print("<---- Incorrect Username or Password ----> Attempts left :", attempts)

if not logged_in:
    print("Access denied. Please contact the agency admin.")
else:
    print("\n---- Login Successful ----\n")
    agency = AgencyList()
    agency.load_from_file()
    agency.display_clients()

    choice = ""
    while choice != "6":
        print("\n----- MENU -----")
        print(" 1. Add New Client")
        print(" 2. Display All Clients")
        print(" 3. Search Client")
        print(" 4. Update Client Package")
        print(" 5. Delete Client")
        print(" 6. Exit")
        choice = input("Enter your choice : ")

    if choice == "1":
        name = input("Client / Business Name : ").replace(" ", " ").strip()
        if name == "":
            print("Name cannot be empty")
        elif agency.find_client(name) is not None:
            print("Client", name, "already exists")
        else:
            phone = input("Phone Number (10 digits) : ")
            while not (phone.isdigit() and len(phone) == 10):
                phone = input("Invalid! Enter 10 digit Phone Number : ")
            print("Select Platform : ")
            platform = choose("Platform number : ", PLATFORMS)
            print("Select Service : ")
            service = choose("Service number : ", SERVICES)
            budget = ask_budget()
            agency.add_client(name, phone, platform, service, budget)
            agency.save_to_file()
            print("<---- Client stored successfully in clients.txt ---->")

```

```
elif choice == "2":
    print("----- ALL CLIENTS -----")
    agency.display_clients()

elif choice == "3":
    key = input("Enter Client Name or Phone Number to search : ")
    agency.search_client(key)

elif choice == "4":
    key = input("Enter Client Name or Phone Number to update : ")
    if agency.find_client(key) is None:
        print("Sorry!", key, "is not available in the client list")
    else:
        print("Select New Service :")
        service = choose("Service number : ", SERVICES)
        budget = ask_budget()
        if agency.update_client(key, service, budget):
            agency.save_to_file()

elif choice == "5":
    key = input("Enter Client Name or Phone Number to delete : ")
    if agency.delete_client(key):
        agency.save_to_file()

elif choice == "6":
    print("----- THANK YOU FOR USING THE AGENCY SYSTEM -----")

else:
    print("Please enter a valid option (1-6)")
```

OUTPUT:-

Login and Client List

The first password is wrong. The second attempt matches a line in login.txt, and the clients are loaded from clients.txt into the linked list.

```
=====
      SOCIAL MEDIA MARKETING AGENCY SYSTEM
=====
Hello! Welcome --- What's your name : Riya
Riya, please enter Username : admin
Riya, please enter Password : admin12
<---- Incorrect Username or Password ----> Attempts left : 2
Riya, please enter Username : admin
Riya, please enter Password : admin123

---- Login Successful ----
```

NO	NAME	PHONE	PLATFORM	SERVICE	BUDGET
1	TrendWave Apparel	9822334455	Instagram	Reels Editing	12000
2	Patil Sweets	9876543210	Facebook	Post Design	5000
3	FitZone Gym	9123456780	YouTube	Ad Campaign	18000

```
-----
Total Clients : 3      Total Monthly Budget : Rs. 35000
```

1. Add New Client

The phone number is checked for 10 digits. The new node is added at the end of the list and the file is saved.

```
----- MENU -----
1. Add New Client
2. Display All Clients
3. Search Client
4. Update Client Package
5. Delete Client
6. Exit
Enter your choice : 1
Client / Business Name : Cafe Aroma
Phone Number (10 digits) : 98765
Invalid! Enter 10 digit Phone Number : 9000011111
Select Platform :
  1. Instagram
  2. Facebook
  3. YouTube
  4. LinkedIn
Platform number : 1
Select Service :
  1. Post Design
  2. Reels Editing
  3. Ad Campaign
  4. Account Management
Service number : 3
Monthly Budget (Rs.) : 8000
<---- Client stored successfully in clients.txt ---->
```

2. Display All Clients

The list is traversed from the head; the new client is last.

Enter your choice : 2

----- ALL CLIENTS -----

NO	NAME	PHONE	PLATFORM	SERVICE	BUDGET
1	TrendWave Apparel	9822334455	Instagram	Reels Editing	12000
2	Patil Sweets	9876543210	Facebook	Post Design	5000
3	FitZone Gym	9123456780	YouTube	Ad Campaign	18000
4	Cafe Aroma	9000011111	Instagram	Ad Campaign	8000

Total Clients : 4 Total Monthly Budget : Rs. 43000

3. Search Client

Linear search by name – found and not found.

Enter your choice : 3

Enter Client Name or Phone Number to search : FitZone Gym

Client Found!

Name : FitZone Gym

Phone : 9123456780

Platform : YouTube

Service : Ad Campaign

Budget : Rs. 18000

Enter your choice : 3

Enter Client Name or Phone Number to search : Green Leaf

Sorry! Green Leaf is not available in the client list

4. Update Client Package

The client is found by phone number and the service and budget are changed.

Enter your choice : 4

Enter Client Name or Phone Number to update : 9876543210

Select New Service :

1. Post Design
2. Reels Editing
3. Ad Campaign
4. Account Management

Service number : 4

Monthly Budget (Rs.) : 15000

Patil Sweets updated successfully!

5. Delete Client

TrendWave Apparel was the head node, so the head now points to the next node.

Enter your choice : 5

Enter Client Name or Phone Number to delete : TrendWave Apparel

TrendWave Apparel deleted successfully!

2. Display All Clients (after update and delete)

Enter your choice : 2

----- ALL CLIENTS -----

NO	NAME	PHONE	PLATFORM	SERVICE	BUDGET
1	Patil Sweets	9876543210	Facebook	Account Management	15000
2	FitZone Gym	9123456780	YouTube	Ad Campaign	18000
3	Cafe Aroma	9000011111	Instagram	Ad Campaign	8000

Total Clients : 3 Total Monthly Budget : Rs. 41000

6. Exit

```
Enter your choice : 6
----- THANK YOU FOR USING THE AGENCY SYSTEM -----
```

clients.txt after the run

The file always matches the linked list because it is saved after every change.

```
Patil Sweets,9876543210,Facebook,Account Management,15000
FitZone Gym,9123456780,YouTube,Ad Campaign,18000
Cafe Aroma,9000011111,Instagram,Ad Campaign,8000
```

CONCLUSION

The Social Media Marketing Agency System shows how a linked list and file handling in Python can be used to solve a real business problem. The agency can add, display, search, update and delete client records quickly, and the data stays saved in clients.txt between runs. The SWOT analysis shows that social media marketing is a fast-growing field with many opportunities, but also high competition and a strong dependence on platforms. Keeping client records organised and adapting quickly to change are therefore key to an agency's success.